

Exercise 2.11

A survival model follows Makeham's law, so that

$$\mu_x = A + Bc^x, \quad x \geq 0.$$

(a) Show that under Makeham's law

$${}_t p_x = s^t g^{c^x(c^t - 1)},$$

where $s = e^{-A}$ and $g = \exp\{-B/\log c\}$.

(b) Suppose you are given the values of ${}_{10}p_{50}$, ${}_{10}p_{60}$, and ${}_{10}p_{70}$. Show that

$$c = \left(\frac{\log({}_{10}p_{70}) - \log({}_{10}p_{60})}{\log({}_{10}p_{60}) - \log({}_{10}p_{50})} \right)^{0.1}.$$

Solution

(a)

For a life aged x , the survival probability for a further t years is

$${}_t p_x = \exp \left\{ - \int_x^{x+t} \mu_r dr \right\}.$$

Under Makeham's law,

$$\mu_r = A + Bc^r.$$

Therefore,

$$\begin{aligned} \int_x^{x+t} \mu_r dr &= \int_x^{x+t} (A + Bc^r) dr \\ &= \int_x^{x+t} A dr + B \int_x^{x+t} c^r dr \\ &= At + B \left[\frac{c^r}{\log c} \right]_x^{x+t} \\ &= At + \frac{B}{\log c} (c^{x+t} - c^x) \\ &= At + \frac{B}{\log c} c^x (c^t - 1). \end{aligned}$$

Thus,

$$\begin{aligned} {}_t p_x &= \exp \left\{ -At - \frac{B}{\log c} c^x (c^t - 1) \right\} \\ &= e^{-At} \exp \left\{ -\frac{B}{\log c} c^x (c^t - 1) \right\}. \end{aligned}$$

Now let

$$s = e^{-A} \quad \text{and} \quad g = \exp \left\{ -\frac{B}{\log c} \right\}.$$

Then

$$e^{-At} = s^t$$

and

$$\exp \left\{ -\frac{B}{\log c} c^x (c^t - 1) \right\} = g^{c^x (c^t - 1)}.$$

Therefore,

$$\boxed{{}_t p_x = s^t g^{c^x (c^t - 1)}}.$$

(b)

From part (a),

$${}_t p_x = s^t g^{c^x (c^t - 1)}.$$

Taking logarithms gives

$$\log({}_t p_x) = t \log s + c^x (c^t - 1) \log g.$$

For $t = 10$, this becomes

$$\log({}_{10} p_x) = 10 \log s + c^x (c^{10} - 1) \log g.$$

Hence,

$$\log({}_{10} p_{50}) = 10 \log s + c^{50} (c^{10} - 1) \log g,$$

$$\log({}_{10} p_{60}) = 10 \log s + c^{60} (c^{10} - 1) \log g,$$

$$\log({}_{10} p_{70}) = 10 \log s + c^{70} (c^{10} - 1) \log g.$$

Subtracting consecutive equations eliminates the term $10 \log s$:

$$\log({}_{10} p_{70}) - \log({}_{10} p_{60}) = (c^{70} - c^{60}) (c^{10} - 1) \log g,$$

$$\log({}_{10} p_{60}) - \log({}_{10} p_{50}) = (c^{60} - c^{50}) (c^{10} - 1) \log g.$$

Taking the ratio gives

$$\begin{aligned} \frac{\log({}_{10} p_{70}) - \log({}_{10} p_{60})}{\log({}_{10} p_{60}) - \log({}_{10} p_{50})} &= \frac{(c^{70} - c^{60}) (c^{10} - 1) \log g}{(c^{60} - c^{50}) (c^{10} - 1) \log g} \\ &= \frac{c^{60} (c^{10} - 1)}{c^{50} (c^{10} - 1)} \\ &= c^{10}. \end{aligned}$$

Therefore,

$$c^{10} = \frac{\log({}_{10} p_{70}) - \log({}_{10} p_{60})}{\log({}_{10} p_{60}) - \log({}_{10} p_{50})}.$$

Taking the tenth root,

$$\boxed{c = \left(\frac{\log({}_{10} p_{70}) - \log({}_{10} p_{60})}{\log({}_{10} p_{60}) - \log({}_{10} p_{50})} \right)^{0.1}}.$$